

The Chain Rule — From Calc I to Calc III

In Calc I, the chain rule handled **compositions** of single-variable functions:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Example: If $y = \sin(x^2)$, let $u = x^2$. Then $y = \sin u$, so:

$$\frac{dy}{dx} = \cos(u) \cdot 2x = 2x \cos(x^2)$$

Now in Calc III: What if z depends on x and y , and x and y each depend on other variables?

There are **multiple paths** from the independent variable to z — we need to account for **all of them**.

Calc I chain rule: one path, one product. Calc III chain rule: **multiply along each branch, add across branches**.

Case 1: One Independent Variable

Suppose $z = f(x, y)$ where $x = g(t)$ and $y = h(t)$. Then z is ultimately a function of t alone.

Chain Rule (Case 1)

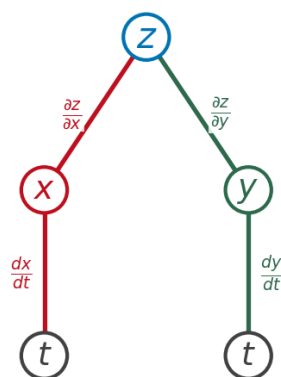
If $z = f(x, y)$ is differentiable, and $x = g(t)$, $y = h(t)$ are differentiable functions of t , then:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Notice the pattern:

- One term for **each intermediate variable** (x and y)
- Each term is a product: (partial of z) $\times$ (derivative of that variable)
- The terms are **added** together

A **variable tree** makes the pattern visual. Draw the dependencies, then read off the formula:



How to read the tree:

1. Each **branch** from top to bottom is a path from z to t
2. **Multiply** the derivatives along each branch
3. **Add** the products from all branches

Left branch: $\frac{\partial z}{\partial x} \cdot \frac{dx}{dt}$

Right branch: $\frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$

Total: $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$

Draw the tree **first**, then read off the formula. **Multiply along each path, add across paths**.

Example 1: One Independent Variable

Example 1

Let $z = x^2 + y^2 + xy$, where $x = \sin t$ and $y = e^t$.

Find $\frac{dz}{dt}$ at $t = \frac{\pi}{4}$.

Draw the variable tree, write the chain rule formula, compute each piece, then evaluate at $t = \pi/4$.

Chain rule formula:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Compute the pieces:

$$\frac{\partial z}{\partial x} = 2x + y, \quad \frac{dx}{dt} = \cos t, \quad \frac{\partial z}{\partial y} = 2y + x, \quad \frac{dy}{dt} = e^t$$

Assemble:

$$\frac{dz}{dt} = (2x + y) \cos t + (2y + x) e^t$$

Evaluate at $t = \pi/4$: $x = \frac{\sqrt{2}}{2}$, $y = e^{\pi/4}$

$$\left. \frac{dz}{dt} \right|_{t=\pi/4} = \left(\sqrt{2} + e^{\pi/4} \right) \frac{\sqrt{2}}{2} + \left(2e^{\pi/4} + \frac{\sqrt{2}}{2} \right) e^{\pi/4} = 1 + \sqrt{2} e^{\pi/4} + 2e^{\pi/2}$$

Your Turn: One Variable Chain Rule

Example 2

Let $z = e^x \sin y$, where $x = t^2$ and $y = 3t$.

Find $\frac{dz}{dt}$.

Draw the variable tree, compute the partials and derivatives, then combine.

Chain rule:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Compute:

$$\frac{\partial z}{\partial x} = e^x \sin y, \quad \frac{dx}{dt} = 2t, \quad \frac{\partial z}{\partial y} = e^x \cos y, \quad \frac{dy}{dt} = 3$$

Assemble:

$$\frac{dz}{dt} = 2t e^{t^2} \sin(3t) + 3e^{t^2} \cos(3t) = e^{t^2} [2t \sin(3t) + 3 \cos(3t)]$$

Case 2: Two Independent Variables

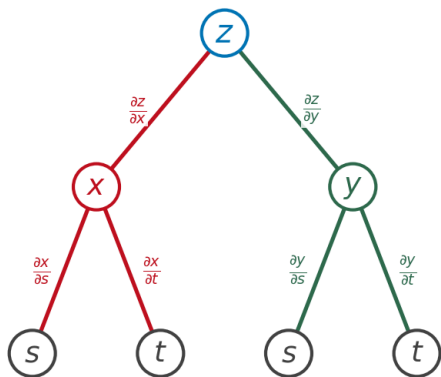
Now suppose $z = f(x, y)$ where $x = g(s, t)$ and $y = h(s, t)$. Then z depends on **two** independent variables s and t .

Chain Rule (Case 2)

If $z = f(x, y)$, $x = g(s, t)$, $y = h(s, t)$, all differentiable, then:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$



The tree now has **four branches** — two paths to s and two paths to t .

For $\partial z/\partial s$: follow only the branches that end at s , multiply along each, and add.

Same idea — but now each intermediate variable (x, y) depends on **both** s and t .

With two independent variables, we get **two formulas** — one for $\partial z/\partial s$ and one for $\partial z/\partial t$. Each follows the same tree pattern.

Example 3: Two Independent Variables

Example 3

Let $z = x^2y^3$, where $x = s \cos t$ and $y = s \sin t$.

Find $\frac{\partial z}{\partial s}$ when $(s, t) = (2\sqrt{2}, \pi/4)$.

Draw the variable tree for $\partial z/\partial s$. Compute partials, then evaluate at the given point.

Chain rule for $\partial z/\partial s$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

Compute the pieces:

$$\frac{\partial z}{\partial x} = 2xy^3, \quad \frac{\partial x}{\partial s} = \cos t, \quad \frac{\partial z}{\partial y} = 3x^2y^2, \quad \frac{\partial y}{\partial s} = \sin t$$

Assemble: $\frac{\partial z}{\partial s} = 2xy^3 \cos t + 3x^2y^2 \sin t$

Evaluate at $(s, t) = (2\sqrt{2}, \pi/4)$: $x = 2\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 2$, $y = 2\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 2$

$$\frac{\partial z}{\partial s} = 2(2)(8) \frac{\sqrt{2}}{2} + 3(4)(4) \frac{\sqrt{2}}{2} = 16\sqrt{2} + 24\sqrt{2} = 40\sqrt{2}$$

The General Chain Rule

The pattern extends to **any** number of intermediate and independent variables.

General Chain Rule

If $u = f(x_1, x_2, \dots, x_n)$ where each $x_i = g_i(t_1, t_2, \dots, t_m)$, then:

$$\frac{\partial u}{\partial t_j} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_j} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_j} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_j}$$

for each $j = 1, 2, \dots, m$.

The recipe is always the same:

1. Draw the variable tree
2. For each path from u down to t_j : multiply the derivatives along the path
3. Add all the path-products together

You don't need to memorize separate formulas for each case. Just **draw the tree** and read off the answer.

Implicit Differentiation Revisited

In Calc I, we used implicit differentiation to find dy/dx when y was defined implicitly by an equation like $x^2 + y^2 = 25$.

In Calc III, we often have an equation like $F(x, y, z) = 0$ that defines z implicitly as a function of x and y . How do we find $\partial z / \partial x$?

Method 1 — Direct differentiation:

- Treat y as a constant and z as a function of x
- Differentiate the equation with respect to x
- Solve for $\partial z / \partial x$

Method 2 — The shortcut formula:

If $F(x, y, z) = 0$ defines z implicitly as a function of x and y , and $F_z \neq 0$:

Implicit Differentiation Formulas

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

Method 2 is derived **from** the chain rule. It's a convenient shortcut — but you should understand why it works.

Why Does $\partial z / \partial x = -F_x / F_z$ Work?

Start with $F(x, y, z) = 0$, where $z = z(x, y)$ is implicitly defined.

Differentiate both sides with respect to x (holding y constant):

By the chain rule:

$$\frac{\partial F}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0$$

Since x is independent: $\frac{\partial x}{\partial x} = 1$. Since y is held constant: $\frac{\partial y}{\partial x} = 0$.

$$F_x \cdot 1 + F_y \cdot 0 + F_z \cdot \frac{\partial z}{\partial x} = 0$$

Solve for $\partial z / \partial x$:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \square$$

The formula comes directly from the chain rule applied to $F(x, y, z) = 0$. Same idea gives $\partial z / \partial y = -F_y / F_z$.

Example 4: Implicit Differentiation

Example 4

The equation $x^2 \sin(yz) + 3z = y^2$ defines z implicitly as a function of x and y .

Find $\frac{\partial z}{\partial x}$ using both methods.

Method 1: Differentiate directly w.r.t. x (y constant, $z = z(x, y)$). Method 2: Set $F = 0$, use $\partial z / \partial x = -F_x / F_z$.

Method 1 — Direct differentiation. Differentiate w.r.t. x (y constant, $z = z(x, y)$):

$$2x \sin(yz) + x^2 \cos(yz) \cdot y \frac{\partial z}{\partial x} + 3 \frac{\partial z}{\partial x} = 0$$

Solve for $\partial z / \partial x$:

$$\frac{\partial z}{\partial x} [x^2 y \cos(yz) + 3] = -2x \sin(yz) \quad \Rightarrow \quad \frac{\partial z}{\partial x} = \frac{-2x \sin(yz)}{x^2 y \cos(yz) + 3}$$

Method 2 — Formula. Set $F(x, y, z) = x^2 \sin(yz) + 3z - y^2 = 0$.

$$F_x = 2x \sin(yz), \quad F_z = x^2 y \cos(yz) + 3$$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{-2x \sin(yz)}{x^2 y \cos(yz) + 3} \quad \checkmark$$

Both methods give the **same answer**. Method 2 is faster once you're comfortable with it.

Your Turn: Implicit Differentiation

Example 5

The equation $x^3 + y^3 + z^3 + 6xyz = 1$ defines z implicitly as a function of x and y .

Find $\frac{\partial z}{\partial x}$ using the formula $\partial z / \partial x = -F_x / F_z$.

Set $F(x, y, z) = 0$, compute F_x and F_z , then apply the formula.

Set up: $F(x, y, z) = x^3 + y^3 + z^3 + 6xyz - 1 = 0$

Partial derivatives: $F_x = 3x^2 + 6yz$, $F_z = 3z^2 + 6xy$

Apply the formula:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{3x^2 + 6yz}{3z^2 + 6xy} = -\frac{x^2 + 2yz}{z^2 + 2xy}$$

Homework

Section 14.5: 5, 7, 13, 15, 17, 19, 25, 35, 37, 39, 44, 49